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## Abstract

**Background:** Artificial intelligence-based clinical decision support systems (AI-CDSS) have the potential to help primary care physicians provide higher quality care with rising case complexity and resource constraints. However, evidence on their effectiveness and safety in real-world primary care practice is lacking.

**Objective:** To synthesize evidence on the real-world effectiveness and safety of AI-CDSS compared with usual care or non–AI-based systems in primary care settings.

**Methods:** We conducted a systematic review following the Cochrane Handbook for Systematic Reviews of Interventions and reported the results according to PRISMA 2020. An information specialist designed search strategies and systematically reviewed Medline, Embase, CINAHL, Web of Science, and CENTRAL. Reviewers independently performed the selection and extraction processes. Data extraction was informed by the DECIDE-AI and CONSORT-AI reporting guidelines, and the APPRAISE-AI tool. We independently assessed risk of bias using RoB-2 or ROBINS-I. Due to heterogeneity, we conducted a narrative synthesis across clinician-, patient-, and system-level outcomes as well as safety information.

**Results:** We identified 4085 records and selected 14 records on 11 distinct studies. We identified 11 AI-CDSS across four experimental (high risk of bias) and seven quasi-experimental studies (serious to critical risk of bias). All studies were conducted in high-income countries. Follow-up ranged from one to 12 months. Machine-learning and deep-learning approaches were the most common (n=8/11). The interventions fulfilled different decision-support functions, most frequently treatment or order facilitation (n=8/11). Clinician-level outcomes were most frequently reported (n=8/11), followed by system-level (n=6/11) and patient-level outcomes (n=5/11). Clinician-level findings were mixed, with several studies reporting reductions in administrative or alert burden. System-level outcomes were heterogeneous and sparsely reported, with isolated studies suggesting improvements in cost-effectiveness, service utilization, patient engagement, program completion, and care delivery efficiency. Patient-level effects were limited or inconsistent, with improvements mainly observed in well-defined conditions with established diagnostic and treatment pathways. Safety evaluation was rare (n=2/11) and limited to technical malfunctions or self-reported side effects.

**Conclusions:** AI-CDSS implemented in primary care demonstrate limited and inconsistent effectiveness at clinician-, patient-, and system-level. Safety evaluation relied mainly on self-reported side effects and technical malfunctions, without structured monitoring for AI-related risks. These findings highlight a gap between the pace of innovation and the readiness of healthcare systems for reliable clinical use. Closing this gap will require workflow-integrated system design, timely and standardized evaluation, and active safety monitoring to support trustworthy implementation in primary care.

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# Original Manuscript

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**Conclusion.** AI-CDSS implemented in primary care demonstrate limited and inconsistent effectiveness at clinician-, patient-, and system-level. Safety evaluation relied mainly on self-reported side effects and technical malfunctions, without structured monitoring for AI-related risks. These findings highlight a gap between the pace of innovation and the readiness of healthcare systems for reliable clinical use. Closing this gap will require workflow-integrated system design, timely and standardized evaluation, and active safety monitoring to support trustworthy implementation in primary care.

**Keywords**: Artificial intelligence, Clinical decision support systems, Primary health care, Systematic review, Patient safety, Effectiveness, Implementation

## INTRODUCTION

Health systems with high-performing primary care services achieve better population health outcomes, greater equity, and lower avoidable hospitalisation rates[1,2]. Primary care is undergoing a structural crisis[3] as providers are faced with an increase in case complexity, a shrinking workforce, and fragmented information systems[4-7]. The exponential growth of relevant scientific evidence affecting medical practice is contributing to this problem, making it increasingly difficult to remain aligned with all clinical best practices[8]. In response to escalating system pressures, artificial intelligence (AI) technologies are rapidly being integrated into primary care practices to support clinicians' workflow[9].

Clinical decision support systems (CDSS) constitute a promising use case for AI, as their regular deployment show an average effect size of 5% on the desired behavior[10]. CDSS assist clinicians in making patient-specific decisions at the point of care by combining data acquisition, processing, and synthesis to generate tailored recommendations[11]. They can provide various forms of support, including alerts, reminders, and access to relevant clinical information[11]. AI-based CDS (AI-CDSS) systems include computational methods to improve classical static, rule-based logic systems toward adaptive, data-driven decision support that enhances precision and personalization[12]. Many AI-CDSS have demonstrated strong technical performance in development, validation, and methodological studies, but evidence of real-world clinical effectiveness in primary care is insufficiently prepare for clinical implementation[4,13].

Generative AI accelerated the pace of deployment and widen research–implementation gap, with evaluated models rarely deployed and deployed systems insufficiently evaluated [14,15]. Recent reviews show that most AI-CDSS studies remain early-stage, small-scale, or methodologically limited, offering little guidance for implementation in primary care[16-19]. The limited applied clinical evidence raised concerns about unintended consequences of AI-CDSS[15], including diagnostic errors[20], overreliance[20], workflow disruptions[21], and inequitable performance across patient groups[22]. We aimed to synthesize the effectiveness and safety evidence of AI-CDSS compared with standard care or non-AI systems in primary care settings.

## METHODS

*Study design*

We conducted a systematic review following the Cochrane Handbook for Systematic Reviews of Interventions[23] and reported the results according to the PRISMA-2020[24]. We registered the protocol in PROSPERO (CRD420251119225) and reported methodological specifications in **Multimedia appendix 1.**

*Eligibility criteria*

We present the eligibility criteria according to the Population, Intervention, Comparator, and Outcomes (PICO) framework[25] in **Table 1**.

**Table 1:** Eligibility criteria based on the PICO framework.

| **Category** | **Eligibility criteria** |
|---|---|
| Population | Health care professionals (physicians, nurses, allied health) or clinical teams.<br><br>Exclusion: Only students. |
| Intervention | AI-based clinical decision support (CDSS).<br><br>CDSS are commonly defined as "computer systems designed to impact clinician decision making about individual patients at the point in time that these decisions are made"[11]. These systems typically involve three main components: (1) acquisition of patient data, (2) data synthesis, and (3) recommendation of a course of action[11]. They encompass a range of functionalities, including alerts, reminders, order sets, drug-dose calculators, care dashboards, and information retrieval systems at the point of care[11]. When CDSS incorporate artificial intelligence (AI) methods such as machine learning, they are commonly referred to as AI-based CDS (AI-CDSS).<br>Exclusion: Presence of co-interventions (i.e., additional interventions delivered alongside the AI-CDSS). |
| Comparators | Standard care with or without traditional (non-AI) CDSS. |
| Outcomes | Any quantitative outcomes at the patient- (e.g., mortality, readmission), clinician- (e.g., adherence, satisfaction), or system-level (e.g., cost, workflow).<br>Adverse outcomes and unintended consequences. |
| Study designs | Randomized controlled trials, non-randomized controlled trials, mixed method studies, prospective cohort studies, and retrospective cohort studies with AI-CDSS clinical integration.<br><br>Exclusion: Retrospective cohort studies without AI-CDSS clinical integration, reviews, editorial, conference abstract, preprints, and protocols. |

| | |
|---|---|
| Setting | Primary care setting. |
| Language of publication | English and French. |

*Information sources and search strategy*

We developed a search strategy in collaboration with an experienced librarian (F.B.) and tested the sensitivity of the strategy with sources that should be identified with the strategy. There was no date restriction. The search strategy was conducted on June 27th, 2025 and is available in **Multimedia appendix 2**. We undertook a comprehensive search across the following electronic databases: Medline (OVID), Embase (Elsevier), CINAHL, Web of Science, and CENTRAL. We performed backward and forward citation searching[26] of the included studies and the literature reviews found during the selection process as well as trial registration checking on August 6th, 2025 to screen for relevant studies.

*Selection and data collection processes*

We used the management software Covidence (Melbourne, Australia) to eliminate duplicates, with the automated function, and to facilitate study selection. We conducted a calibration training on the first 100 references identified to calibrate eligibility criteria interpretation between reviewers (F.N., M.S., S.O.). If inter-rater agreement (percent agreement) was below 80%[27], we clarified the eligibility criteria through discussion and think aloud approach[28] and conducted a new training. Pairs of reviewers (F.N., M.S., S.O.) applied eligibility criteria to independently screen the titles and abstracts of all potentially eligible citations and then to full-text articles retrieved. We resolved discrepancies by team consensus. We documented reasons for full text exclusion and reported them using PRISMA-2020 flowchart.

*Data items*

We developed a standardized data extraction form informed by existing reporting guidelines, including DECIDE-AI for early-stage clinical evaluation of AI-CDSS[29], CONSORT-AI for clinical trials of AI interventions[30], and the APPRAISE-AI tool for quantitative evaluation of AI-CDSS studies[31]. Given the low number of eligible studies, the form was pilot tested on the most detailed study[32]. Two independent reviewers (F.N., S.O.) performed the data extraction and disagreements were resolved by a third one (M.S.). Extracted variables covered study characteristics (e.g., study period, number of participating institutions), participant characteristics (e.g., sample size, baseline features), and intervention details (e.g., purpose, outputs). We also collected data on outcomes and effects (e.g., measurement instruments, between-group comparisons) and safety (e.g., risk mitigation strategies, reported harms or unintended effects).

*Risk of bias assessment*

We assessed individually randomized and cluster randomized trials using the corresponding versions of RoB-2[33] and quasi-experimental studies using ROBINS-I (version 2)[34]. Two reviewers (F.N., S.O.) independently evaluated risk of bias for the main outcome of each study, and disagreements were resolved by team consensus.

*Synthesis method*

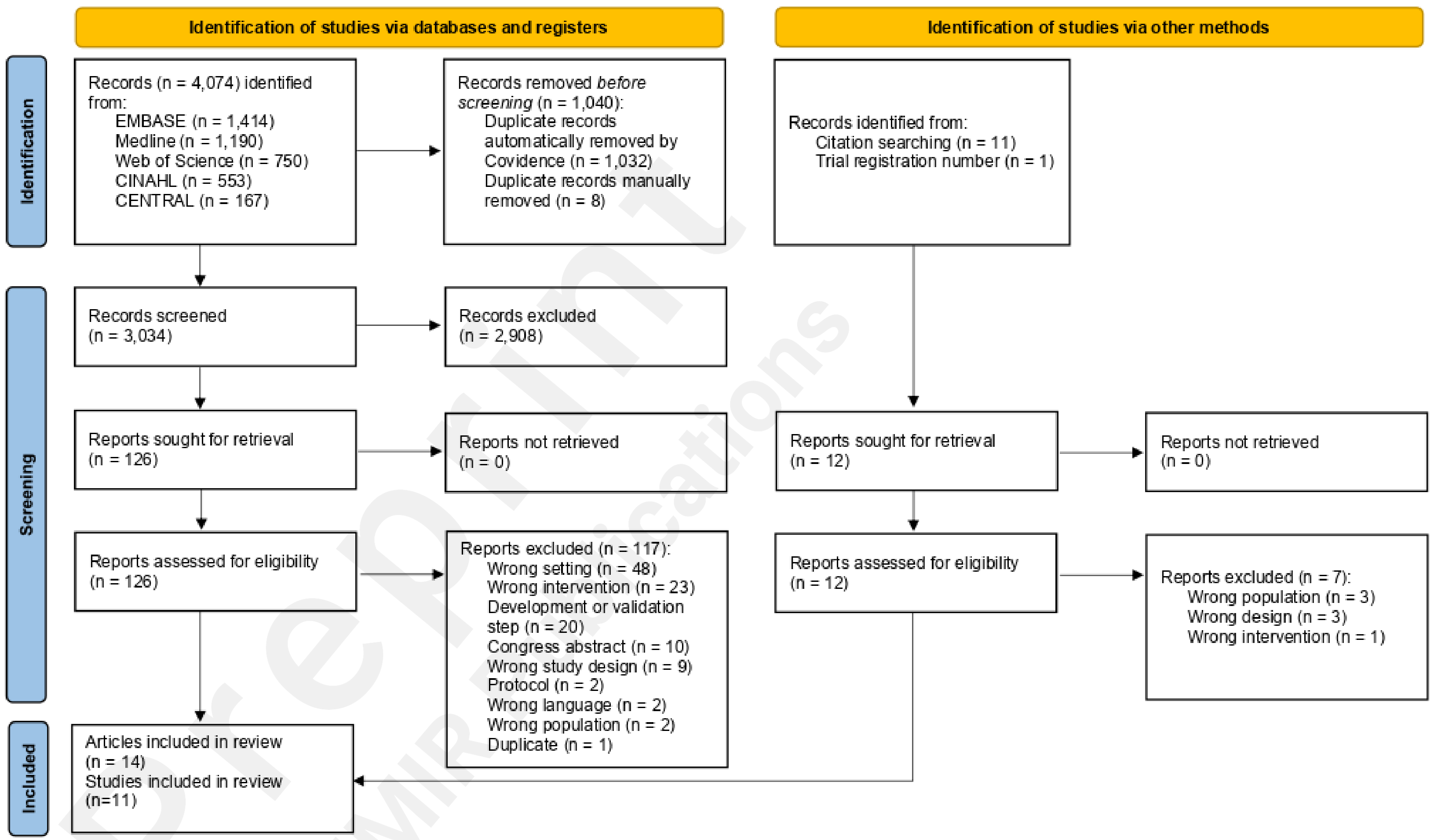
Because of the interventional, clinical, and methodological heterogeneity across studies, we did not conduct a meta-analysis or other quantitative synthesis[35,36]. We performed a narrative synthesis, organizing the findings by outcome level and identifying patterns across studies, while also considering safety evaluation. We categorized outcomes in three levels: patient-level, referring to measures directly affecting patients such as symptom improvement or function; clinician-level, including outcomes related to provider knowledge, behavior, or workload; and system-level, encompassing organizational or workflow outcomes such as healthcare utilization or cost-effectiveness.

## RESULTS

*Study selection*

We retrieved 4,074 records from the search. After removing duplicates, 3,034 records remained for the selection. Of these records, 126 were retained for full-text screening. We identified 12 supplemental records from handsearching and trial registration checking. At the end of the process, we included 14 records on 11 distinct studies. We present a flow diagram of the selection process in **Figure 1**.

**Figure 1:** Flow diagram based on PRISMA-2020.



*Characteristics of the studies*

We present a summary of the characteristics of the included studies in **Table 2** and a detailed version is available in **Multimedia appendix 3**.

The 11 included studies were conducted across North America (n=7, 63.6%) and Europe (n=4, 36.4%). Four studies (36.4%) used experimental designs, and seven (63.6%) were quasi-experimental, applying a variety of comparison strategies including concurrent control groups, historical or matched controls, and before–after or self-controlled analyses. Follow-up durations ranged from approximately one to 12 months. When reported (n=8, 72.7%), the number of participating institutions varied from one to 45 and included urban, rural, or mixed health-care settings. Each study evaluated a distinct AI-CDSS. From the DECIDE-AI reporting items, none of the studies reported patient involvement in the evaluation of the interventions.

**Table 2:** Characteristics of the studies.

| Authors<br>*Year of publication*<br>Country of recruitment | Level of evidence<br>*Control group* | Study period | Number of participation institutions<br>*Setting* | Presence of conflicts of interest | Name of the AI-CDS |
|---|---|---|---|---|---|
| Alcoceba-Herrero et al.[37]<br>*2024*<br>Spain | Experimental study<br>*Usual care* | ~2 months | 25<br>*Urban and rural* | No | No name |
| Areias et al.[38]*<br>*2024*<br>USA | Quasi-experimental study with concurrent control group<br>*Digital care program* | Experimental: ~6 months<br><br>Control: ~24 months | Not reported | Yes | No name |
| Chen et al.[39]<br>*2020*<br>USA | Quasi-experimental study with alternating on/off periods<br>*Usual care* | ~2 months | Not reported<br>*Urban academic* | No | SmartCDS |
| Cruz et al.[40]<br>*2019*<br>Spain | Quasi-experimental before–after study<br>*Usual care* | Before: ~16 months<br><br>After: ~1 month | 24<br>*Not reported* | Not reported | No name |
| Granviken et al.[32]<br>*2025*<br>Norway | Experimental study<br>*Usual care* | ~9 months | 33<br>*Private* | No | SupportPrim PT CDS |
| Herter et al.[41]<br>2022<br>The Netherlands | Quasi-experimental study with matched control group<br>*Usual care* | ~4 months | 36 (experimental) and 29 (control)<br>*Not reported* | Yes | No name |

| | | | | | |
|---|---|---|---|---|---|
| Liu et al.[42]<br>*2021*<br>USA | Quasi-experimental study with historical control group<br>*Usual care* | ~8 months | 1<br>*Academic* | Yes | EyeArt v2.0 ARIAS |
| Popescu et al.[43]<br>*2021*<br>Canada<br>Qassim et al.[44]<br>*2023*<br>Canada | Quasi-experimental study with self-control<br>*Usual care* | ~10 months | Not reported<br>*University hospitals, primary care clinics, and psychiatric clinics* | Yes | Aifred |
| Romero-Brufau et al.[45]<br>*2020*<br>USA | Quasi-experimental before–after study<br>*Usual care* | Before: ~1 month<br>After: ~1 month | 3<br>*Not reported* | No | No name |
| Seol et al.[46]<br>*2021*<br>USA | Experimental study<br>*Usual care* | ~12 months | 1<br>*Not reported* | No | No name |
| Yao et al.[47]<br>*2021*<br>USA<br>Rushlow et al.[48]*<br>*2022*<br>USA<br>Thao et al.[49]*<br>*2024*<br>USA | Experimental study<br>*Usual care* | ~8 months | 45<br>*Academic, community and rural* | Yes | No name |

*Post-hoc analysis

***ARIAS***: Automated Retinal Image Analysis System, ***CDS***: Clinical Decision Support, ***PT***: Physiotherapy, ***USA***: United States of America

*Risk of bias assessment*

We judged the overall risk of bias as high for all experimental studies and as ranging from serious to critical for quasi-experimental designs. We provide overall judgment and detailed domain-level assessments in **Multimedia appendix 4**.

*Characteristics of the participants*

We present a summary of the characteristics of the participants included in the studies in **Multimedia appendix 5**.

The included studies applied AI CDSS to both acute and chronic conditions. Six studies (54.5%) reported the number of recruiting clinicians, which ranged from 7 to 358. Eight studies (72.7%) reported patient recruitment, with experimental-group sample sizes ranging from 14 to 24,083 and control groups from 14 to 12,103 participants. Race or ethnicity was reported in five studies (45.4%); in most of these, participants were predominantly White, with proportions ranging from 15.0% to 94.4%.

*Characteristics of the interventions*

We present AI-CDSS characteristics in **Table 3** and the full description of the experimental and control interventions in **Multimedia appendix 6**.

The included studies reported multiple modes of AI-CDSS delivery, including dashboards (n=5/11, 45.4%), alert-based systems (n=3/11, 27.3%), printed or periodic reports (n=2/11, 18.2%), and email notifications (n=1/11, 9.1%); with one study in which the authors did not specify. The interventions fulfilled different decision-support functions, most frequently treatment or order assistance (n=8/11, 72.7%), followed by risk prediction (n=5/11, 45.4%), patient monitoring (n=3/11, 27.3%), diagnostic support (n=2/11, 18.2%), and alert optimization (n=2/11, 18.2%), with several systems combining multiple functions. All systems requiring a final decision supported, rather than replaced, clinician's judgment. In three studies (27.3%), this was stated explicitly, and in the remainder, clinician control was only implicit. In seven studies (63.6%), clinicians had to perform supplementary tasks to use the AI-CDSS, including manually entering patient characteristics, accessing separate dashboards or portals, and reviewing a report. Four studies (36.4%) reported providing clinician training before integration of the AI-CDSS, and one (9.1%) offered ongoing clinical support during implementation. Machine-learning and deep-learning approaches were the most common (n=8/11, 72.7%). Interoperability with electronic health records was reported in three studies (27.3%), either fully (n=2) or partially (n=1), while the remaining studies implemented standalone systems.

**Table 3:** AI-CDS characteristics.

| Authors | Functions | Delivery type | AI system | Interoperability | Additional tasks | Final decision made by | Training before using AI-CDS |
|---|---|---|---|---|---|---|---|
| Alcoceba-Herrero et al.[37] | ·Alert optimization<br>·Monitoring/follow-up<br>·Documentation/care summary | ·Dashboard<br>·Alerts | Not reported | Standalone | ·Review dashboard | Clinicians* | Technical, logistical, and clinical support |
| Areias et al.[38] | ·Treatment/order facilitation<br>·Monitoring/follow-up<br>·Documentation/care summary | ·Dashboard | Machine learning | Standalone | ·Review and navigate dashboard | Clinicians | No information |
| Chen et al.[39] | ·Alert optimization | ·Alerts | Machine learning | EHR-embedded | ·No | n/a | No information |
| Cruz et al.[40] | ·Treatment/order facilitation | ·Alerts | ·Natural language processing<br>·Machine learning<br>·Deep learning | Partial | ·No | Clinicians* | No information |
| Granviken et al.[32] | ·Treatment/order facilitation<br>·Documentation/care summary | ·Dashboard | Case-based reasoning | Standalone | ·Review and navigate dashboard | Clinicians* | Yes |
| Herter et al.[41] | ·Risk prediction<br>·Treatment/order facilitation<br>·Documentation/care summary | ·Dashboard | Machine learning | Standalone | ·Enter patient's data<br>·Review bar chart | Clinicians* | Yes |
| Liu et al.[42] | ·Diagnosis/screening | ·No information | Deep learning | Standalone | ·Capture fundus images (assumed) | Clinicians* | No |
| Popescu et al.[43]<br>Qassim et al.[44] | ·Risk prediction<br>·Treatment/order facilitation<br>·Monitoring/follow-up | ·Dashboard | Deep learning | Standalone | ·Review and navigate dashboard | Clinicians | Yes |
| Romero-Brufau et al.[45] | ·Risk prediction<br>·Treatment/order facilitation | ·Printed report | Not reported | Standalone | ·Clinical assistant print the report<br>·Review the report | Clinicians | No information |
| Seol et al.[46] | ·Risk prediction<br>·Treatment/order | ·Quarterly report | ·Machine learning<br>·Natural language | Standalone | No | Clinicians* | No information |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | facilitation<br>·Documentation/care summary | | processing | | | | |
| Rushlow et al.[48]<br>Thao et al.[49]<br>Yao et al.[47] | ·Risk prediction<br>·Diagnosis/screening<br>·Treatment/order facilitation | ·Email alerts | ·Deep learning | EHR-embedded | No | Clinicians* | Website access with videos and FAQs |

* assumed as (not explicitly reported); ***FAQs***: Frequently Asked Questions ; ***n/a***: not applicable

*Outcomes and effects*

We present a summary of between-groups effects in **Table 4** and provide a detailed table, including within-group results, in **Multimedia appendix 7**.

Among the 11 included studies, outcomes were most frequently reported at the clinician level (8/11, 72.7%), followed by the system level (6/11, 54.5%) and the patient level (5/11, 45.4%). Nearly half of the studies (5/11, 45.4%) assessed outcomes across all three levels.

At the clinician-level, effects were mixed. Experimental trials tended to report more favorable results than quasi-experimental designs. The most consistent benefits involved reductions in administrative and alert burden without reducing care quality. Effects on prescribing behavior and guideline adherence were heterogeneous. Attitudinal outcomes suggested perceived improvements in coordination and patient preparation, alongside persistent concerns about utility and professional roles.

At the patient-level, both experimental and quasi-experimental studies showed limited or inconsistent effects. Pain intensity and mental health outcomes were the only outcomes evaluated across more than one study. AI-CDSS implementation appeared more effective in acute or well-defined conditions with established diagnostic and treatment decisions, such as urinary tract infections and low ejection fraction (n=2/11, 18.2%). For studies addressing complex, chronic conditions such as musculoskeletal pain and asthma (n=3/11, 27.3%), they all reported neutral results, occasionally with minor adverse effects. Improvements in patient satisfaction were not accompanied by improvements in symptom burden or mental health outcomes.

At the system level, findings were heterogeneous and often reported in single studies. Two studies showed favorable effects: improved cost-effectiveness and increased echocardiography use among high-risk patients; greater program completion, increased patient-initiated messaging, increased physiotherapist reach-outs, and improved physiotherapist-to-patient ratios among people living with chronic musculoskeletal pain.

**Table 4**: Between groups effects for outcomes with p-value.

| Authors | Purpose | Clinician-level effects | Patient-level effects | System-level effects |
|---|---|---|---|---|
| **Experimental studies** | | | | |
| Alcoceba-Herrero et al.[37] | Support early detection of patient deterioration in COVID-19 home isolation | + Reduction of alerts | | |
| Granviken et al.[32] | Support personalised decision-making for musculoskeletal pain management | + More treatment modalities | Ø Global perceived effect<br>Ø Pain intensity<br>Ø Pain self-efficacy<br>Ø Emotional distress<br>Ø General MSK health<br>Ø Number of pain sites<br>Ø Health-related Quality of life<br>Ø Sleep<br>Ø Vitality<br>Ø Use of pain medication last week<br>- Functional status<br>- Work ability | Ø Number of consultations |
| Seol et al.[46] | Support asthma management in children | + Reduction of clinician's burden | Ø Occurrence of asthma<br>Ø Asthma control status | Ø Health care costs<br>Ø Timeliness of follow-up after exacerbation |
| Rushlow et al.[48]<br>Thao et al.[49]<br>Yao et al.[47] | Support early detection of low ejection fraction in adults without prior heart failure | | + Newly diagnosed low ejection fraction | + Echocardiography utilization for high-risk patients<br>+ Cost-effectiveness<br>Ø Echocardiography utilization for overall cohort |
| **Quasi experimental studies** | | | | |
| Areias et al.[38] | Support monitoring and adjustment of digital care programs for individuals living with chronic musculoskeletal pain | | + Satisfaction with the program<br>Ø Pain response rate<br>Ø Mental health<br>- Pain intensity | + Completion rate of the program<br>+ Reach-outs<br>+ Number of messages initiated by patients<br>+ PT-to-patient ratios<br>Ø Safety<br>- Sessions performed per week |
| Chen et al.[39] | Reduce alert fatigue and increase the signal-to-noise ratio of shingles vaccination | Ø Order count<br>Ø Interacted alert counts | | |

| | | | | |
|---|---|---|---|---|
| | reminders | | | |
| Cruz et al.[40] | Support guideline adherence for multiple conditions | + Adherence rate to 3 recommendations<br>Ø Adherence rate to 5 recommendations | | |
| Herter et al.[41] | Support selection of the most effective antibiotic for urinary tract infections | +/ø/- Prescription behavior | + Successful treatment | |
| Liu et al.[42] | Support early detection of diabetic retinopathy and improve adherence to follow-up eye care | | | + Rate of attendance at ophthalmology appointments |
| Popescu et al.[43]<br>Qassim et al.[44] | Support treatment selection and management for major depressive disorders | Ø Appointment length | | |
| Romero-Brufau et al.[45] | Support glycemic control in patients with di | Clinician attitudes:<br>+ Care coordination<br>+ Patient preparation for managing diabetes<br>+ Familiarity with AI<br>Ø Reducing complications<br>Ø Intervention effectiveness<br>Ø Right intervention<br>Ø Routinely use AI<br>Ø AI will complicate my job<br>Ø AI will make my job obsolete<br>- Excited about AI<br>- AI does not understand my job | | |

+: favors the experimental group; **ø**: no difference; **–**: favors the control group

***AI***: Artificial Intelligence; ***PT***: Physiotherapist

*Safety evaluation*

Evaluation and reporting of patient safety and system robustness were inconsistent across the 11 studies. Two studies (18.2%) pre-specified structured monitoring of adverse events, which was focused on patient-reported side effects or standardized questionnaires rather than active surveillance of AI-related harms. Risk mitigation strategies were described in four studies (36.4%) and included human-in-the-loop oversight, internal quality checks, and automated safeguards such as validation of severe alerts or continuous retraining with monitoring dashboards. Reports of errors or malfunctions were uncommon (n=3/11, 27.3%) and mainly incidental, involving image-quality issues, software glitches, or technical exclusions of participants. One study (9.1%) explicitly noted the absence of major errors, but systematic quantification of error rates was rarely attempted.

## DISCUSSION

This systematic review synthesizes real-world evidence on the safety and effectiveness of AI-CDS systems in primary care. Across 11 studies implemented in diverse primary care contexts, AI-CDSS showed modest and inconsistent effects on patient-, clinician-, and system-level outcomes. Effect trends suggest reduced clinicians' administrative and alert burden without evidence of adverse effects on patient safety. The strength of effectiveness evidence remains limited, heterogeneous, and prone to methodological bias, reflecting the early stage of translation from technical validation to small-scale integration.

Patient safety is the foundation of clinicians' trust[50] and AI-CDSS adoption[51]. Across the included studies, safety monitoring was rare (n=2/11) and mainly restricted to technical malfunctions or self-reported side effects. This narrow focus offers limited insight into how AI-generated recommendations shape clinical risk assessment in primary care[52]. It aligns with evidence that safety outcomes are infrequently evaluated in AI-CDSS studies[53-56]. A concern involves model–data mismatch. AI models trained on data that fail to reflect the diversity and complexity of primary care populations may produce misleading suggestions[4,57-59]. Clinicians may attempt to mitigate this risk by overriding AI recommendations that diverge from their clinical reasoning. However, automation bias can weaken this safeguard. Automation bias is the tendency of humans to favor machine-generated decisions even when contradicted by clinical judgment or available data[60]. This risk is well documented in experimental CDS studies[61,62], where incorrect automated suggestions are accepted in 6 to 11% of cases[62]. Emerging evidence suggests similar patterns for AI-CDSS[63-66]. Beyond the direct risk of implementing erroneous recommendations, repeated reliance on AI output may contribute to clinician deskilling[63], further compounding safety vulnerabilities. Recognizing these patient safety risks, national regulatory initiatives and AI governance frameworks call for ongoing monitoring of deployed systems[67-70]. Yet they offer limited operational guidance on how to evaluate the questions clinicians confront in practice: when to trust an AI recommendation, when to question it, and how to recognize when it misaligns with the patient in front of them. For AI-CDSS to be trusted, safety evaluations must move beyond detecting malfunctions and examine how these systems affect clinical judgment in the realities of primary care.

Demonstrating effectiveness remains essential for clinicians for successful AI-CDSS adoption in routine care[51,71]. In the studies included, effectiveness was not extensively covered. The most consistent improvements were observed in workflow outcomes, including reduced clinician administrative burden and improved process efficiency. The effects on patient-level outcomes were mixed and often negligible. These patient-level findings align with evaluations in hospital and specialty settings, which also report heterogeneous effects[56]. Benefits appear only in psychiatry, are mixed in oncology and neurology, and show no evidence of effectiveness in anesthesiology[56]. However, in psychiatry, the observed benefits were reported only for low-complexity cases[72], suggesting that case complexity may influence AI-CDSS effects. This pattern parallels our findings, where improvements in patient outcomes appeared mainly in simpler clinical pathways. These pathways involved conditions with a unimodal, guideline-concordant management as selecting antibiotics or clinical actions defined by a straightforward procedural step as ordering an ECG for suspected ventricular dysfunction. When case complexity increases, multiple guideline-concordant management options often coexist, and no single "gold-standard" pathway dominates the decision-making[73]. In these situations, AI-CDSS recommendations may rarely alter choices clinicians would have made independently[74]. For example, Granviken et al. reported that in highly complex conditions as musculoskeletal pain[75-77], clinicians mainly used the AI-CDS to support their current practice rather than to incorporate new alternatives suggested by the system[32]. This limited capacity to influence clinician behavior in complex scenarios and contributes to the limited patient-level effects across AI-CDSS clinical implementation studies. Short follow-up periods offer another explanation for the modest and often negligible patient outcomes. In our review, follow-up duration ranged from one to twelve months, a period during which clinicians were often still learning to integrate the AI-CDSS into their clinical workflow[78]. Such learning curve can affect engagement, trust, and the consistent application of recommendations and limits early improvements in patient outcomes. Given the substantial workload pressures in primary care, improvements in efficiency and workflow support offer immediate value even while longer-term evaluation is needed to assess patient-level effects.

Workflow integration is essential for clinicians to incorporate AI-CDSS into routine care[51,71]. In our review, eight systems were deployed as standalone tools outside the electronic health record. This requires clinicians to manually enter information, navigate separate dashboards, or review printed or digital reports during the encounter. Because these steps fall outside the usual workflow, they introduce additional time costs and cognitive load into already short appointments, reducing the likelihood of sustained us[73,78,79]. Training challenges reinforced this pattern. Four studies offered dedicated training, yet one explicitly reported that more extensive sessions were needed before clinicians could incorporate the system efficiently[32]. This need for training indicates that clinicians must adopt new cognitive and behavioral routines (i.e., digital readiness and capabilities[80,81]) to use the system effectively, routines that are difficult to establish when the tool remains external to the natural flow of clinical work[5,78,82]. Full electronic health record integration, including automatic retrieval of patient data and automated display of recommendations, offers a strategy to mitigate these barriers[74,78]. Only two interventions in our review achieved such integration and both reported positive outcomes. Chen et al.[39] reduced alert counts without compromising vaccination orders, and EAGLE trial[47-49] improved identification of patients with low ejection fraction and increased use of echocardiography among high-risk patients in a cost-effective manner. Although these findings do not establish causality, they suggest that workflow integration may help create conditions for consistent use and observable impact. Reducing the burden associated with AI-CDSS use is therefore essential to support its adoption in routine care.

This review has subject-matter and methodological limitations. Our search strategy was developed in collaboration with an information specialist (F.B.) and tested for sensitivity. However, we identified several studies through citation searching and trial registration number. This reflects the indexing inconsistencies and incomplete database coverage for CDSS, despite the existence of a MeSH term since 1997. Our methodology, including backward and forward citation screening in identified knowledge syntheses records, reduces the risk of missing studies that would affect the overall result interpretation. Substantial heterogeneity across interventions, study designs, and reported outcomes precluded meta-analysis methods, removing summarized effects estimates for the strength-of-evidence reporting. Although narrative data synthesis was appropriate, it carries a risk of interpretative bias. We mitigated this risk through review by a multidisciplinary team of researchers, clinicians, and people with lived experience, strengthening the external validity.

## CONCLUSION

AI-based clinical decision support systems implemented in primary care demonstrate limited and inconsistent effectiveness, with safety poorly assessed. Benefits emerge primarily through reductions in clinician burden, whereas patient- and system-level effects remain weak and safety monitoring is passive. Most systems required additional clinician effort and operated outside the electronic health record, reflecting limited integration into existing workflows. Outcome heterogeneity and incomplete reporting restrict interpretation and cross-study comparison. These findings highlight a gap between the pace of innovation and the readiness of healthcare systems for reliable clinical use. Closing this gap will require workflow-integrated system design, timely and standardized evaluation, and active safety monitoring to support trustworthy implementation in primary care.

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**Conflicts of interest**

None declared.

**Abbreviations**
AI: Artificial intelligence
AI-CDSS: Artificial intelligence-based clinical decision support systems
CDSS: Clinical decision support systems

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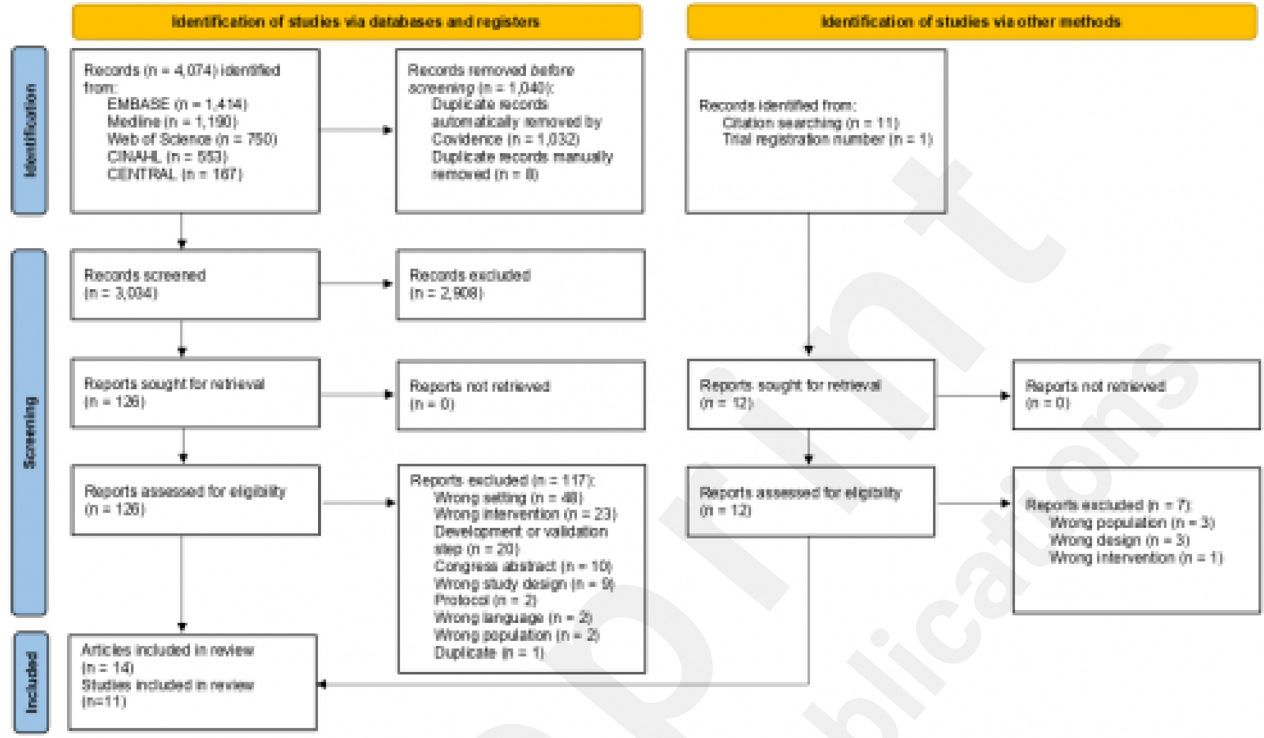
# Supplementary Files

# Figures

Flow diagram based on PRISMA-2020.

**Figure 1:** Flow diagram based on PRISMA-2020.



# Multimedia Appendixes

Methodological changes between report and protocol.
URL: http://asset.jmir.pub/assets/38aa621d63bd560969f2d0f8404f8ba1.docx

Databases search strategies.
URL: http://asset.jmir.pub/assets/dd2a638cf2f8ef129d5bdef1806b1b31.docx

Detailed characteristics of the included studies.
URL: http://asset.jmir.pub/assets/3528e312318d3b5e963b87fc6e75f21d.docx

Risk of bias assessment.
URL: http://asset.jmir.pub/assets/b244cd33774dbac494de3d0752408d3e.docx

Characteristics of the participants.
URL: http://asset.jmir.pub/assets/ab91c70ccddc2fc0f6cde88fbc8b6dda.docx

Characteristics of the interventions.
URL: http://asset.jmir.pub/assets/9891323aecff0a7e0974ddd1e897be91.docx

Outcomes and effects.
URL: http://asset.jmir.pub/assets/bd40f815b13028651d5038dc74594bc6.docx